NextJS Tutorial #1 - Introduction

Concepts : What is NextJS? NextJS vs ReactJS, Use Cases & Course Roadmap

Overview of This Lecture

This is the introduction video of the full NextJS course. In this lecture we understand:

- What **ReactJS** is, what it can do and where it falls short
- What **NextJS** is and what extra features it gives
- Difference between a **Single Page Application (SPA)** and a **Framework**
- Why learn NextJS if we already know React
- Common **use cases** of NextJS
- The **roadmap** of this course

Prerequisites

To learn NextJS comfortably, we should know:

- Basic **HTML, CSS & JavaScript**
- Basic knowledge of **React** (a few introductory React videos will also be covered in this playlist)

> **Note** - Watch each video carefully first, then code along on your own. Practising the code yourself is the best way to learn NextJS.

What is React?

React is a JavaScript **library** used to build **user interfaces (UI)**. Everything visible on the screen - buttons, forms, lists, etc. - can be displayed using React. It is a **front-end** JavaScript library.

What React can do

- Create **components** (header, footer, card, etc.)
- Update the UI **fast**
- Help us write **reusable code**

Limitations of React

- It is **not built-in** (not a complete solution) - we have to write/set up most things ourselves
- It is **not SEO friendly** by default
- It does **not support back-end** work

> **Note** - Want to become a **front-end developer**? React is the right choice. Want to become a **full-stack developer**? Learn NextJS along with React.

What is NextJS?

NextJS is a **framework of React**. It is not a separate JavaScript library - it is built on top of React and uses React itself.

Simple definition:

```
NextJS = React + Routing + SEO + Performance + Server features
```

Extra features NextJS gives (that React does not)

Feature	Meaning
File-based routing	Create a page (file) and routing is ready automatically
SEO friendly	Better Search Engine Optimization, so the site is easier for search engines to understand
Server Side Rendering (SSR)	NextJS supports server-side rendering (React supports client-side rendering)
API routes	Build back-end APIs inside the same project
Production-ready setup	No need to search for a separate back-end; can connect with databases like MongoDB, PostgreSQL

Because of all this, we can become a **full-stack developer** using just one application.

> **Note** - NextJS makes good SEO *easier* through server rendering, but it does not guarantee a #1 Google rank - content quality and SEO practices still matter.

React vs NextJS

React	NextJS
Only a UI library	A framework of React - works on front-end as well as back-end
SEO is weak	SEO is strong
Routing and back-end must be set up separately	Routing and server features are built-in
Supports client-side rendering	Supports server-side rendering (and more)

Conclusion: For real-world projects, NextJS is more practical than plain React.

Single Page Application (SPA) vs Framework

Single Page Application (React)

- Moving from one page to another does **not reload the whole page**
- Only the content that needs to change is updated

Framework (NextJS)

- We can build an even **smoother** experience than a plain SPA
- Also supports **server-side rendering**, so SEO automatically becomes strong

Why NextJS and Not Just React?

Choose NextJS when you need:

Requirement	Choose
Search Engine Optimization	NextJS
Rank the site higher on Google	NextJS
Easy routing	NextJS
Fast performance	NextJS
Production-ready setup	NextJS

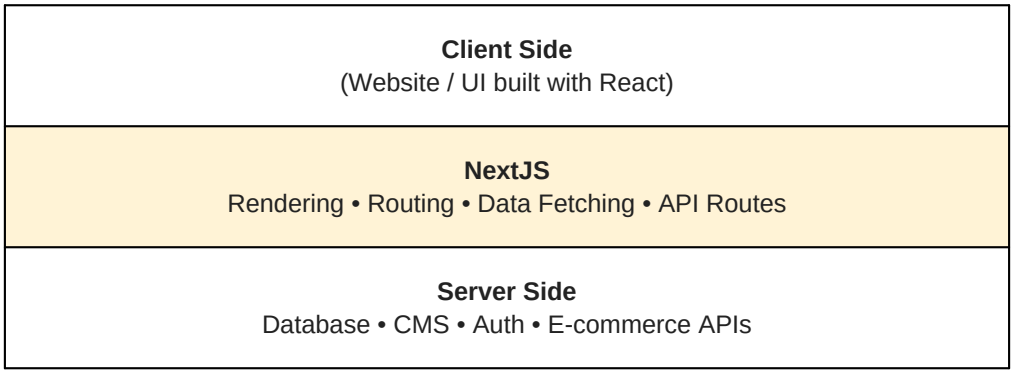
In all these cases NextJS is the clear winner.

Use Cases of NextJS

- Blogs and content-related websites
- E-commerce websites
- Company / business websites
- Dashboards
- Learning platforms

How NextJS Works (Architecture)

NextJS sits between the **client** and the **server** and connects them:



- The UI we build with React is shown on the client side
- **Rendering, routing and data fetching** are all handled by NextJS (even the React UI can be built through NextJS)
- To connect client and server, NextJS manages **routing, APIs, database connection, CMS, Auth and e-commerce APIs**
- So front-end and back-end both are managed by NextJS

> **Note** - The course follows the official **Learn Next.js** course on the NextJS website (link in the video description), explained in an easier way.

Course Roadmap

The official Learn Next.js course is divided into **4 parts**:

Part	Topic
1	React Fundamentals
2	Page Router
3	App Router
4	Search Engine Optimization (SEO)

Rough order of what we will learn in this course:

- React fundamentals and NextJS setup
- Routing and creating pages
- Data fetching
- SEO (Search Engine Optimization)
- Performance improvement
- Deployment

Interview questions related to each topic will also be discussed daily.

Quick Revision - Interview Questions

Q1. What is the difference between React and NextJS?
Ans: React is only a UI library; NextJS is a React framework with built-in routing, SSR, SEO support and back-end features.

Q2. Is NextJS a separate language/library from React?
Ans: No. It is a framework built on top of React.

Q3. What is file-based routing?
Ans: Routes are created automatically from the files/pages we create - no separate router setup needed.

Q4. Which rendering does React use by default and which does NextJS support?
Ans: React: client-side rendering. NextJS: server-side rendering (along with others).

Q5. Why is NextJS better for SEO?
Ans: Pages can be rendered on the server, so search engines receive ready content.

Q6. Where is NextJS commonly used?
Ans: Blogs, e-commerce sites, company websites, dashboards and learning platforms.

Key Takeaways

- React = UI library (front-end). NextJS = React framework (front-end + back-end).
- NextJS = React + Routing + SEO + Performance + Server features.
- NextJS gives file-based routing, SSR, API routes and a production-ready setup.
- Learn NextJS for SEO, easy routing, fast performance and full-stack development.
- Use cases: blogs, e-commerce, company websites, dashboards, learning platforms.

> *Keep Learning & Keep Exploring!*